

Math 170S: Statistics

Spring 2019

Instructor:	Elizaveta (Liza) Rebrova
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CCLE Webpage:	https://ccle.ucla.edu/course/view/19S-MATH170S-1 (used for announcements only; please subscribe to receive them)
Course Webpage:	http://www.math.ucla.edu/~rebrova/170S.html (all crucial course info, syllabus and homework will be here)
Anonymous feedback form:	goo.gl/xNxFNb
Lectures:	
Room	5138 MS
Time	MWF 1:00pm – 1:50pm
Office Hours:	
Room	5338 MS
Hours	TBA (or by appointment)
Teaching Assistant	John Zhang
Discussion session room	5138 MS
Discussion session hours	H 1:00pm – 1:50pm
Office hours	TBA on discussion class

Course description

Lecture, three hours; discussion, one hour. Requisites: courses 31A, 31B, 170E. Introduction to statistics. Topics include sampling, estimation and properties of estimators, and construction of confidence intervals and hypotheses testing.

Course materials

- **Textbook:** Hogg, Tanis, Zimmerman Probability and Statistical Inference, 9th edition
- **Auxiliary textbook:** Introduction to Probability by D. P. Bertsekas and John N. Tsitsiklis, 2nd edition

Grading

Your grade is formed based on 8 homework assignments, one midterm, and the final exam. MyUCLA gradebook will be used for grading. The final score will be computed by the following scheme:

- homeworks - 15%, the midterm - 35%, the final - 50%

Any grading complaints and requests for re-grading must be submitted **within 1 week of receiving your score**. Grading complaints not initiated within this period of time will not be considered.

Homework

- Homework assignments will be issued weekly and posted on the course website <http://www.math.ucla.edu/~rebrova/170S.html>. It is turned in the beginning of each discussion section (and returned on the next discussion section).
- Your score on homework assignment will be based on a few randomly chosen problems, as well as overall completion of the assignment.
- **Late homework will not be accepted.**
- One lowest homework score will be dropped.
- Printed or handwritten homeworks have to be submitted.
- You may use whatever resources you want to do the homework, including computers, textbooks, friends, the TA, etc. Collaboration in small groups (up to 5 people) is encouraged. You can indicate the names of your collaborators if you'd like to. However, **all homework assignments must be written by you, i.e. you cannot copy someone else's solution or submit a joint solution made by several people.**

Exams

- There will be one midterm and one final exam.

Midterm: May 6 (Monday), in class

Final: June 11 (Tuesday), 11:30pm–2:30pm, room TBA

- You **must take the final exam** in order to pass the class.
- No make-up exams will be possible (unless exceptional circumstances, validated by the Undergraduate office).
- Students must bring their UCLA ID cards to the midterms and to the final exam.

Tentative course schedule

This is a *very* tentative schedule. Section numbers refer to the main textbook.

Week	Date	Section	Topics
1-1	M, Apr 1	6.1	Intro to Statistics, histograms
1-2	W, Apr 3	6.1, 6.2	Exploratory Data Analysis
	H	Discussion	
1-3	F, Apr 5	6.2, 6.3	Order Statistics
2-1	M, Apr 8	6.4	Maximum Likelihood Estimation
2-2	W, Apr 10	6.4	Method of Moments Estimation
	H	Discussion	HW 1 collected
2-3	F, Apr 12	6.5	A Simple Regression Problem
3-1	M, Apr 15	6.7	Sufficient Statistics
3-2	W, Apr 17	6.7	Sufficient Statistics
	H	Discussion	HW 2 collected
3-3	F, Apr 19	6.8	Bayesian Estimation
4-1	M, Apr 22	6.8	Bayesian Estimation
4-2	W, Apr 24	7.1	Confidence Interval for Means
	H	Discussion	HW 3 collected
4-3	F, Apr 26	7.2	Confidence Intervals for the Difference of Two Means
5-1	M, Apr 29	7.3	Confidence Interval for Proportions
5-2	W, May 1	7.4	Sample Size
	H	Discussion	HW 4 collected
5-3	F, May 3	7.5	Distribution-Free Confidence Intervals for Percentiles
6-1	M, May 6	6.1-7.5	Midterm
6-2	W, May 8	8.1	Hypothesis testing, Type I and II errors
	H	Discussion	
6-3	F, May 10	8.1	Tests About One Mean
7-1	M, May 13	8.2	Tests of the Equality of Two means
7-2	W, May 15	8.3	Tests About Proportions
	H	Discussion	HW 5 collected
7-3	F, May 17	8.4	The Wilcoxon Test
8-1	M, May 20	8.5	Power of a Statistical Test
8-2	W, May 22	8.6	Best Critical Regions
	H	Discussion	HW 6 collected
8-3	F, May 24	9.1	Chi-Square Goodness-of-Fit Tests
9-1	M, May 27	No class	Memorial Day holiday
9-2	W, May 29	9.2	More Chi-Square tests
	H	Discussion	HW 7 collected
9-3	F, May 31	9.3	One-Factor Analysis of Variance
10-1	M, Jun 3	9.4	Two-Way Analysis of Variance
10-2	W, Jun 5	all	Review
	H	Discussion	HW 8 collected
10-3	F, Jun 7	all	Review